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ADVANCED STUDIES & RESEARCH BOARD

BLOCK –IV, PDA BUILDING, PHASE-V, HAYATABAD

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No: _____

Date: _____

RESEARCH PROPOSAL

TITLE: Knowledge and Practices Regarding Infection Prevention and Control Among Nurses Working in Adult ICUs at Public and Private Hospitals of Peshawar, Pakistan

Name of Candidate: Zia Ullah Jan, Irfan Ullah, Shah Faisal, Muhammad Shayan Khan, Sadia Israr, JanBaz, Asim U Din, Waheed Ullah, Muhammad Hamza, Faisal Mashal Name

of Supervisor: Dr Syed Yousaf Shah (Principal of Northwest College Nursing Peshawar)

CoSupervisors:

- 1. Sir Shoaib Awan (Lecturer at Northwest College Nursing_Peshawar)**
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Duration of Project: 6 Months

Institute: Life Care College of Nursing and Allied Health Sciences Peshawar.

Budget Required: _____

Name & Signature of Student/Scholar: _____

Name & Signature of the Supervisor: _____

Name & Signature of Head of Institute: _____



1.TITLE:

Knowledge and Practices Regarding Infection Prevention and Control Among Nurses Working in ICUs in Public and Private Hospitals of Peshawar, Pakistan.

2.INTRODUCTION:

Background Information: Infection prevention and control (IPC) is a practical, evidence-based strategy aimed at protecting both patients and healthcare workers from preventable infections. Its primary goal is to reduce infection rates through policies and practices that control and minimize the spread of illnesses [1].

Health care-associated infection an infection acquired by a patient during the process of care (including preventive, diagnostic and treatment services) in a hospital or other health care facility, which was not present or incubating at the time of admission. Hospital Acquired Infections (HAIs) are serious public health problems in both industrialized and developing economies, causing great morbidity, mortality and associated with high expenditure [2]. On average, out of every 100 patients in acute- care hospitals, seven patients in high-income countries (HICs) and 15 patients in low- and middle-income countries (LMICs) will acquire at least one HAI during their hospital stay [3]. Healthcare-associated infections (HAIs) continue to pose a significant threat, particularly among patients admitted to intensive care units (ICUs). This elevated risk is primarily attributed to the high prevalence of invasive procedures, immunosuppression, multiple comorbidities, frailty, and advanced age in this population [4]. Almost up to one third (30%) of patients in intensive care can be affected by HAIs, with an incidence that is two to 20 times higher in LMICs than in HICs [5]. In addition, cultural and contextual factors uniquely affect healthcare practices in countries with varying resource levels and healthcare infrastructure, such as Pakistan. The Pakistani healthcare system faces challenges, including limited access to ongoing training for staff, insufficient resources, and systemic healthcare delivery issues, which can increase the risks of HAIs [6]. The themes of IPC compliance, knowledge, and practices are critical for reducing the prevalence of healthcare- associated infections (HAIs), particularly in intensive care units (ICUs).

Notably, nearly 70% of HAI can be prevented by the implementation of effective IPC interventions [7]. However, the compliance with IPC practices has been a critical implication on the implementation of the IPC program. The findings from various researchers shown that the burden of health care acquired infections were associated with poor risk reflection to infection prevention programs, which was not considered by authorities, unhygienic practices [8]. Nurses have frequent and direct contact with the patients due to which, their knowledge and awareness about the IPC practices are integral [9]. Therefore, it's important to examine nurses' current knowledge and practices in Pakistani healthcare sectors to develop effective strategies that align with local needs and health challenges. Given the increase burden of HAIs in Pakistan, this study aims to assess ICUs staff nurses' knowledge and practices regarding IPC.

Problem Statement: Effective IPC is a fundamental component of providing high-quality patient care in ICUs in maintaining patient safety. Nurses play a central role in this process, as they are responsible for effective infection prevention and control practices. in fact, despite of standard guidelines recent study suggests that there is a growing issue that nurses may not have adequate knowledge and adherence to best practices when it comes to actual practice. The problem stems from the potential consequences of ineffective infection control, specifically the ICUs patients which are vulnerable to infections due to prolong length of stay, the extended use of invasive devices (like ETT, Mechanical Ventilation, CVC, Urinary Catheter etc.) which are the significant cause of morbidity and mortality. This research aims to explore the gap between infection control practices and the reality of ICUs nurses' knowledge and practices, ultimately try to improve patient care.

Rationales: Limited evidence is available regarding infection prevention and control practices among ICUs nurses in Peshawar particularly in comparing public and private healthcare sectors. Understanding these dynamics is vital for implementing tailored educational interventions that equip nursing staff with the skills needed to reduce infection effectively. The insights gained from this research will serve as a foundation for improving IPC training, education and ultimately reducing the incidence of HAIs within Pakistani healthcare settings, contributing to better patient health outcomes.



Furthermore, it is important to identify and understand the current state of ICUs nurses' knowledge and practices regarding effective infection prevention and control practices to identify areas of improvement, and propose solutions to enhance the infection prevention control measures quality in healthcare settings.

3.OBJECTIVE(S):

Objectives of this Study Are:

1. To Assess the Knowledge and Practice of ICUs Nurses about IPC in Public and Private Hospital.
- 2.To Compare the knowledge and practice score of Public and private hospital nurses.
- 3.To correlate the knowledge level with practice level regarding Infection prevention and control.

4. OPERATIONAL DEFINITIONS:

IPC: Infection prevention and control (IPC) is a clinical and public health specialty and a set of measures, based on a practical, evidence-based approach. The aim of IPC is to prevent patients, health workers from preventable infection.

Nurse: Person Completed a program of basic generalized education, licenced by Pakistan nursing council and have more than 1-year experience in intensive care unit.

Knowledge: is defined as the conscious awareness and understanding of information, facts, or skills about IPC by assessed on adapted questionnaire about knowledge using a 20-item scale. Level of Knowledge was classified as poor (<50%), moderate (50–79%), and good (80–100%).

Practice: is the extent to which nurses' practices are per the recommended WHO guidelines. Evaluating nurses action by an adapted questionnaire. We assess the practice of infection control standard precautions using 15 questions. The overall level of practice was classified as poor (<50% score), moderate (50–79%), and good (80–100%).

5. HYPOTHESIS:

H1: There is a significant difference in knowledge and practice scores of public and private hospitals nurses.

H1: There is positive correlation between knowledge and practice scores regarding infection prevention and control among ICUs nurses.

6. MATERIALS AND METHODS:

6a. Study Design: Quantitative Analytical cross sectional Study

6b. Study Settings: Public and Private Tertiary Care Hospitals in Peshawar (KTH, HMC, RMI, PGH)

6c. Study Duration: 6 Months

6d. Sample Size: we calculated Sample size on Rao soft calculator total population are 170 on 95% confidence level, 5% margin of error the resulted calculated sample size is :119



6e. Sampling Technique: Non Probability (convenience Sampling)

7a. Inclusion Criteria:

- ☐ Registered nurses working for at least one year in the selected Adult intensive care unit.

7b. Exclusion Criteria:

- Nurses who are unwilling to participate
- Nursing director
- Nursing students
- Nursing Internes

8. DATA COLLECTION PROCEDURE:

We will obtain Ethical approval from ERB, and then we will take permission from hospital administration and signing informed consent form. After getting permission and signing informed consent form from administration, we will take informed consent form from participants also if they are agreeing to participate voluntarily. Participants will have informed about the study's goals and objectives prior to completing the questionnaires, which included a written explanation. Participation will be voluntary, and participants will have assured that they could choose not to participate without any consequences. To ensure anonymity, no identifying information such as names or hospital details will have collected, and unique codes will be assigned to participants to maintain confidentiality. We will assess the knowledge and practice of infection prevention and control using an adapted structured questionnaire. The questionnaire will be divided into three parts: the first part included questions on demographic and professional information of the Nurses; the second and third parts respectively focused on the knowledge and practices regarding infection prevention and control. [10]

We will assess knowledge using a 20-item scale. We provided 1 point for each right answer and 0 points for each wrong answer. The maximum possible score will be 20 points with a range from 0 to 20 points.

We will assess the practice of infection prevention and control using 15 questions on practicing standard precautions. We will be give 1 point for each activity they were always practicing and 0 points for not practicing. The maximum possible score was 15 points with a range from 0 to 15 points.



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9.DATA ANALYSIS PROCEDURE:

This section explains how we will look closely at our data to find important things. Our goal is to have a well-structured dataset that's free from errors or inconsistencies, making it suitable for meaning full analysis. We will use descriptive statistics to summarize and present key features of our dataset. Like Frequencies and percentages will be calculated for categorical variables such as gender, educational qualification, hospital type, and years of experience. Continuous variables like (Knowledge and Practice score) will be summarized using the mean and standard deviation or the median. We will use inferential statistics just like Independent t-test will be used to compare mean knowledge and practice scores between Public and private hospital nurses and Apply Correlational test for to know the relationship between Knowledge and practice. We will use visual aids like tables, pie charts, graphs and description to illustrate relationships, distributions and trends with in our dataset, enhancing our understanding of the research.

We have chosen to employ SPSS Version 26.0.

10.GANTT CHART:

S. No.	ACTIVITIES	Month 1				Month 2				Month 3	Month 4	Month 5
		W 1	W2	W3	W4	W1	W2	W3	W4	W1-4	W1-4	W1-4
1.	Proposal development											
2.	Submission of proposal for ethical approval											
3.	Data collection											
4.	Data entry & analysis											
5.	Report writing											
6.	Total time	2 weeks		1 weeks		5 weeks				2 months		1month



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11. BUDGET:

ITEM	UNIT COST	QUANTITY	TOTAL COST
Printing of questionnaires	Rs.15	200 copies	Rs.3,000
Stationery & office supplies (pens, files, folders)	Rs.1,000	Lump sum	Rs.1,000
Data entry & SPSS analysis	Rs.6,000	Lump sum	Rs.6,000
Report printing & binding	Rs.2,000	Lump sum	Rs.2,000
Data collection travel (multiple Hospitals involved)	—	Lump sum	Rs.5,000
Miscellaneous / contingency	Rs.2,000	Lump sum	Rs.2,000
GRAND TOTAL			Rs.22,000

10.BIBLIOGRAPHY:

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10. Abalkhail A, Al Imam MH, Elmosaad YM, Jaber MF, Al Hosis K, Alhumaydhi FA, et al. Knowledge, attitude and practice of standard infection control precautions among health-care workers in a university hospital in Qassim, Saudi Arabia: A cross-sectional survey. Int J Environ Res Public Health. 2021;18(22):11831. <https://dx.doi.org/10.3390/ijerph182211831>

ANNEXE;

Annexure I; Informed Consent Form

Annexure II; Data collection Instrument Attached

INFORMED CONSENT FORM

Study Title:

Knowledge and Practices Regarding Infection Prevention and Control among Nurses Working in Adult Intensive Care Units of Public and Private Hospitals in Peshawar, Pakistan

Investigators:

Zia Ullah Jan, Irfan Ullah, Shah Faisal Khan, Muhammad Shayan Khan, Sadia Israr, JanBaz, Asim U Din, Waheed Ullah, Muhammad Hamza, Faisal Mashal.

Institution: Life Care College of Nursing and Allied Health Sciences Peshawar.

Contact Information:

Email: ziakmu2022@gmail.com

Purpose of the Study:

You are invited to participate in a research study that aims to assess the knowledge and practices regarding Infection Prevention and Control (IPC) among nurses working in Adult Intensive Care Units (ICUs) of public and private hospitals in Peshawar, Pakistan. The findings of this study will help identify existing knowledge and practice gaps and may contribute to improving infection prevention strategies, nursing education, patient safety, and the quality of healthcare services.

Procedures:

If you agree to participate, you will be asked to complete a structured questionnaire. The questionnaire will take approximately 10–15 minutes to complete. There are no right or wrong answers. You are requested to answer each question honestly based on your current knowledge and routine clinical practice.



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Risks and Benefits:

There is no known physical, psychological, or social risks associated with participation in this study. Although you may not receive any direct personal benefit, your participation will contribute to research that may help improve Infection Prevention and Control practices, reduce healthcare-associated infections, and enhance the quality of nursing care in Intensive Care Units.

Confidentiality:

All information collected during this study will be kept strictly confidential. No personal identifiers, such as your name or employee identification number, will be collected. The information provided will be used solely for research purposes and will be reported only in summarized form. Your responses will remain anonymous and accessible only to the research team.

Voluntary Participation:

Participation in this study is completely voluntary. You have the right to refuse participation or withdraw from the study at any time without any penalty, consequences, or effect on your employment or professional standing.

Consent Statement:

By signing below, you confirm that you have read and understood the information provided above. You have had the opportunity to ask questions, and all your questions have been answered satisfactorily. You voluntarily agree to participate in this research study.

Participant's Name: _____

Participant's Signature: _____

Date: _____



QUESTIONNAIRE FOR IPC KNOWLEDGE AND PRACTICE

SECTION 1: DEMOGRAPHIC INFORMATION

1. Gender

☐ Male

☐ Female

2. Age

☐ 22-34 years

☐ >34 years

3.Exposed to infection while working

☐ Yes

☐ No

4.Received training on infection control

☐ Yes

☐ No

5.Duration in Current ICUs Unit/Department

☐ 0–2 years

☐ 2–4 years

☐ 5–10 years

☐ >10 years

6.Hospital You Current Practicing as a Nurse

☐ Government

☐ Private



SECTION II: KNOWLEDGE REGARDING INFECTION PREVENTION AND CONTROL

Knowledge Statement	True	False
1. Standard precautions are used for the care of all patients regardless of their diagnosis and perceived infection status.		
2. Isolation precaution is one of the elements in standard precaution.		
3. Washing hands after contact with the patient's environment is one of the elements in standard precaution.		
4. Alcohol-based rubs are used after removing gloves.		
5. Performing hand hygiene is required before and after patient care.		
6. Hands should be washed with soap and water before and after handling potentially infectious materials irrespective of wearing gloves.		
7. PPE is important in infection control because it acts as a barrier between infectious materials such as viral and bacterial contaminants and your skin, mouth, nose, or eyes (mucous membranes).		
8. Gloves must be worn every time during handling potentially infectious materials.		
9. Gloves must be changed during patient care if you move hands from 'contaminated body site' to 'clean body site'.		
10. Surgical masks can protect the nose and mouth when procedures and activities are likely to generate splashes or sprays of blood and body fluids.		
11. The purpose of using a gown or apron is to protect clothes from splashes or sprays of blood and body fluids.		
12. Remove all personal protective equipment (PPE) before leaving the patient's environment.		
13. Stationery, telephones kept in wards, and doorknobs can be sources of infections.		
14. All linen from an infectious patient should be thrown in a red linen bag even when it is free from visible blood or body fluids.		



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15. Segregation of clinical and non-clinical waste is important for preventing the spread of infection.		
16. Ampoules injection that has been used must be disposed of in the clinical waste bin.		
17. Recapping of needles, in general, is not appropriate.		
18. If you puncture hand with sharp instruments, you must report to the concerned authorities.		
19. Puncture-proof containers should be used for disposal of sharps objects.		
20. Mask must be placed on coughing patients to prevent potential dissemination of infectious respiratory secretions from the patient to others.		

SECTION III: PRACTICES REGARDING INFECTION PREVENTION AND CONTROL

Practice Statement	Always Practicing	Not Practicing
1. Always performs hand hygiene when they come in contact with patients.		
2. Always performs hand hygiene after taking off gloves.		
3. Always washes hands immediately after contacting any blood, body fluid, secretion, excretion, or dirty substances.		
4. Always wears gloves when drawing blood samples.		
5. Always wears gloves when disposing of stool or urine.		
6. Always wears gloves when handling impaired patient skin.		
7. Always wears gloves when handling the patient's mucosa.		
8. Always wears gloves when handling saliva or sputum culture.		



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9. Always wears gloves when performing parenteral injections of medications.		
10. Always wears gloves when dressing wounds.		
11. Always wears gloves when they come in contact with blood.		
12. Always wears mask when performing operations/procedures that might induce spraying of blood, body fluid, secretions, or excretions.		
13. Always wears a protective eye patch or goggle when performing operations/procedures that might induce spraying of blood, body fluid, secretions, or excretions.		
14. Always wears protective suits or gown when performing operations/procedures that might induce spraying of blood, body fluid, secretions, or excretions.		
15. Always disposes of needles, blades, or any other single-use sharp objects in a sharp disposal container after use.		

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